Eli McPherson
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Sr. Biomedical Engineer | Lead ML Engineer in Transcriptomics/Proteomics

Summary

Senior biomedical engineer with ten years of coding, computational biology, and in-lab biology experience, five years of leadership experience overseeing teams of 2-40 people. In my current work, I balance two roles for small, stealth-stage Boston biotech startups: as lead ML engineer, I oversee development of ML biomolecule feature prediction software with a team of two other engineers; as lead proteomics engineer, I lead development and design of bioinformatics analysis pipelines for early-stage drug development.

Skills

Programming Languages: Python, MATLAB, R
ML tools: PyTorch, TensorFlow, Scikit-learn, Biopython, OR-Tools, Ray, ESM
Workplace Collaboration and Leadership: overseeing teams of 2-40 people, project management, translating between cross-functional teams, ML/DevOps, Jira, high-throughput automation integration, SOP generation and validation, Jira, verbal and written presentation of results
Bioinformatics tools: PyMOL, HMMER, BLAST, KEGG, GO, STRING, HISAT2, STAR, cellranger, MultiQC, Nextflow, Snakemake, nf-core
Cloud & infrastructure: Git, Docker**,** SQL, HPC workflows, AWS**,** Azure**,** GCP
In-Lab Biological Automation Instrumentation: Hamilton STAR, Opentrons, Echo, Biostack, Synergy H1, JESS
Laboratory & Experimental Techniques: DNA Extraction, Chromatography, Assay Development, Strain Design, Directed Evolution, ELISA, Fluorescence/Luminescence Assays, High Throughput Lab Automation, Drug Discovery

Experience

Lead ML Engineer | Stealth-Stage Startup | Boston, MA | November 2024 – present
- Oversee development of custom NN architecture for generation of targeted protein characteristics in ML biomolecule prediction software, with two other engineers.
- Architect and build scalable code repositories using Ray to augment model tuning and training across GPUs, and production code repositories for automated generation and evaluation of candidate biomolecules. These contributions cut biomolecule generation from weeks to hours, and increased biomolecule evaluation by 1000%.

Lead Bioinformatics Engineer | Stealth-Stage Startup | Boston, MA | August 2024 – present
- Propose, design, and implement bioinformatics drug target identification pipeline using PDB, GO, and STRING.
- Oversee development of assay analysis pipeline, implementing large scRNAseq and bulkRNAseq dataset analyses using Nextflow and associated preprocessing and differential expression analysis.

Student Researcher | Khalil Lab, Boston University | Boston, MA | September 2023 – August 2024
- Developed custom Python modules integrating ML algorithms for luminescence detection in DIY bioreactors (eVOLVER), resulting in successfully defended MS thesis, and robust characterization of this methodology's specific limiting factors.
- Independently led experimental design for this project and collaborated closely with Khalil lab members to integrate findings into lab's eVOLVER platform.

Senior Research Associate | Resonance Medicine | Boston, MA | January 2023 – March 2023
- Scaled a directed evolution platform from manual bench-based assays to a semi-automated, high-throughput workflow that reduced hands-on experimental time by 75%, directly resulting in improved experimental precision and associated reductions in cost for company's drug discovery pipeline.

Research Associate I/II | Joyn Bio | Boston, MA | September 2020 – October 2022

- Independently proposed and implemented an automated high-throughput robotics workflow for liquid handlers, achieving a 400% improvement in experimental throughput and 90% reduction in manual intervention.
- Developed and implemented quantitative immunoassay protocols for detection of toxic molecules in host bacterial organisms. Independently conducted and validated in vitro data analysis and communicated verbal and written results to cross-functional teams (Strain Design, Bioengineering, Leadership).
- Acted as primary liaison with external collaborator to translate results of in vitro immunoassay protocols to in vivo assay protocols, resulting in improved product efficacy validation.
- Wrote, reviewed, and validated SOP protocols for integration of laboratory automation in close collaboration with interdisciplinary teams, resulting in fundamental changes to the company's assay procedure.

Research Assistant | Colorado State University | Fort Collins, CO | August 2014 – December 2017

- Developed novel protocols for loading guest molecules into computationally designed protein hosts, and established chemical conjugation workflows to stabilize protein crystals, resulting in two publications.

Other Experience

Founder & CEO, Paper Crane Coffee Roasters (Nov 2022 – Present)

- Entrepreneurial leadership experience managing operations, partnerships with local Boston, MA businesses, and product quality.

Peace Corps Volunteer – Ethiopia (Jan 2018 – Mar 2020)

- Provided community health education initiatives.

United States Marine Corps – Sergeant (Aug 2006 – Jul 2010)

- Leadership and team coordination of 40 Marines, and mission operations management. Specialized in IT/satellite communications infrastructure. Deployed to Iraq.

Education

Master of Science in Biomedical Engineering – Boston University, 2024
Bachelor of Science in Chemical and Biological Engineering – Colorado State University, 2017

Publications

McPherson, E. (2024). Luminescence detection in eVOLVER for autonomous bioreactor control. Boston University M.S. Thesis. https://hdl.handle.net/2144/49235

Huber, T. R., **McPherson, E. C.,** Keating, C. E., & Snow, C. D. (2018). Installing Guest Molecules at Specific Sites within Scaffold Protein Crystals. Bioconjugate Chemistry, 29(1), 17–22. https://doi.org/10.1021/acs.bioconjchem.7b00668

Huber, T. R., Hartje, L. F., **McPherson, E. C.,** Kowalski, A. E., & Snow, C. D. (2017). Programmed Assembly of Host–Guest Protein Crystals. Small (Weinheim an Der Bergstrasse, Germany), 13(7), np-n/a. https://doi.org/10.1002/smll.201602703